

FAQ Assistant API

Project Planning Phase Document

Item	Details
Project Name	FAQ Assistant API
Document	Project Planning
Version	1.0 (Draft)
Phase	Phase 1 - Planning
Status	For review

1. Project Overview

The FAQ Assistant API is a backend service that answers users' frequently asked questions automatically. A client application (website, mobile app or chatbot) sends a question to the API, and the API returns the most relevant answer from a managed FAQ knowledge base. Administrators can add, edit and remove FAQs without changing any code.

2. Problem Statement

Support teams answer the same questions again and again, which wastes time and delays replies. Users also struggle to find answers in long static FAQ pages. The project solves this by giving instant, consistent answers through a simple, reusable API.

3. Objectives

- Provide a REST API that returns accurate answers to user questions in real time.
- Allow admins to manage FAQ content (create, read, update, delete) securely.
- Support matching of similar or differently-worded questions, not only exact text.
- Keep the design simple, documented and easy to integrate with any client.
- Track basic usage so unanswered questions can be identified and added later.

4. Scope

4.1 In Scope

- Question-answer endpoint (ask a question, get the best matching answer).
- FAQ management endpoints (CRUD) with categories and tags.
- Search and ranking of FAQs by relevance with a confidence score.
- Authentication using API keys / tokens for admin operations.
- Logging of queries and unanswered questions.
- API documentation (OpenAPI / Swagger) and a basic test suite.

4.2 Out of Scope (for this version)

- Full front-end or admin dashboard UI.
- Voice interface and multi-language translation.
- Live-agent handover and ticketing integration.
- Advanced analytics dashboards.

5. Key Features and Deliverables

Feature	Description	Priority
Ask endpoint	POST /ask - takes a question, returns best answer and confidence score	High

Feature	Description	Priority
FAQ management	CRUD endpoints for FAQs, categories and tags	High
Smart matching	Keyword and semantic similarity to match reworded questions	High
Authentication	API key / token based access, admin-only write operations	High
Fallback handling	Default reply and logging when no good match is found	Medium
Usage logging	Store questions, matches and feedback (helpful / not helpful)	Medium
API documentation	Auto-generated OpenAPI docs with examples	Medium
Rate limiting	Prevent misuse and control load	Low

6. Stakeholders and Roles

Role	Responsibility
Project Owner / Guide	Approves scope, reviews milestones and final deliverables
Project Manager / Team Lead	Planning, task allocation, progress tracking
Backend Developer(s)	API design, implementation, database, matching logic
Tester / QA	Test plan, test cases, bug reporting
End Users / Client Apps	Consume the API to show answers to their users
Admin	Maintains FAQ content in the knowledge base

7. Proposed Technology Stack

The stack below is a suggestion and can be changed based on the team's skills and project requirements.

Layer	Proposed Choice
Language / Framework	Python with FastAPI (or Node.js with Express)
Database	PostgreSQL / MySQL (or MongoDB) for FAQs and logs
Matching / NLP	TF-IDF or sentence embeddings for similarity search
Authentication	API keys or JWT tokens
Documentation	OpenAPI / Swagger UI, Postman collection
Testing	Pytest / Jest, Postman for API testing
Version Control	Git and GitHub
Deployment	Docker, cloud host (Render / AWS / Azure)

8. Project Phases and Timeline

The project follows the phases below. Durations are estimates and can be adjusted.

Phase	Main Activities	Output	Duration
1. Planning	Define scope, objectives, risks, schedule	Project Planning Document	1 week
2. Requirements	Gather functional and non-functional requirements	SRS Document	1 week
3. Design	Architecture, API design, database design	Design Document, ER diagram	1-2 weeks
4. Development	Implement endpoints, matching logic, auth	Working API code	3 weeks
5. Testing	Unit, integration and API testing, bug fixing	Test Plan, Test Report	1-2 weeks
6. Deployment	Deploy, configure, prepare user guide	Live API, User Guide	1 week
7. Maintenance	Monitor, fix issues, add new FAQs / features	Maintenance Log	Ongoing

9. Milestones

Milestone	Completion Criteria
M1 - Plan approved	Planning document reviewed and signed off
M2 - Requirements frozen	SRS approved, no open requirement questions
M3 - Design complete	API contract and database schema finalized
M4 - First working build	Ask and FAQ management endpoints working end to end
M5 - Testing complete	All critical test cases passed, major bugs fixed
M6 - Release	API deployed and documentation published

10. Risk Analysis

Risk	Impact	Mitigation
Poor answer accuracy for reworded questions	High	Use semantic matching, add confidence threshold and fallback reply
Scope creep	Medium	Freeze scope after requirements; handle new ideas as future versions
Incomplete or low-quality FAQ data	High	Collect and review FAQ content early; provide admin tools to improve it
Security issues (unauthorized edits)	High	Token authentication, input validation, role-based access
Schedule delays	Medium	Weekly progress reviews, buffer time in each phase

Risk	Impact	Mitigation
Performance under heavy load	Medium	Indexing, caching, rate limiting, load testing

11. Assumptions and Constraints

- An initial set of FAQ content will be available from the project owner.
- Client applications will handle their own user interface.
- The team has access to required tools, hosting and a database.
- The project must be completed within the agreed timeline and available resources.

12. Success Criteria

- The API returns a relevant answer for most in-scope questions with fast response time.
- Admins can manage FAQs without developer help.
- All high-priority features are delivered, tested and documented.
- The API can be integrated by a new client using only the documentation.

13. Next Phase

After approval of this document, the project moves to Phase 2 - Requirements, where detailed functional and non-functional requirements will be prepared in the SRS document.